

Machine Learning End Semester Project Evaluation

Suicidal Sentiment Identifier

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Abstract: Motivation



- Suicidal behavior claims **over 700,000 lives annually**, making it a leading cause of death worldwide.
It is the **fourth most common cause of death** among young people aged **15-29 years**.
- Social media and messaging platforms provide valuable insights into early suicidal behavior and mental health.
- Develop a machine learning model to **detect/flag suicidal tendencies** in text-based communication (social media posts, messages).
- Analyze text data from social media posts and messages to assess potential risk.
- Use a classification model to identify **high-risk users** by analyzing their speech patterns and the context of their communication.
- Create a powerful tool for early detection and intervention in **life-threatening behaviors**, potentially saving lives.

Detecting and Analyzing Suicidal Ideation on Social Media Using Deep Learning and Machine Learning Models

[Theyazn H H Aldhyani](#)^{1,*}, [Saleh Nagi Alsubari](#)²
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9565132/>

Online suicide prevention through optimized text classification

Dr. P. Golda Jeyasheeli[1], C. Kamaleshwar[2], K.Sakthi Aswin[2]
https://www.researchgate.net/publication/323065338_Online_suicide_prevention_through_optimised_text_classification

Predicting depression via social media

Chris Poulin, Brian Shiner, Paul Thompson, Linas Vepstas, Yinong Young-Xu, Benjamin Goertzel, Bradley Watts, Laura Flashman, Thomas McAllister
https://www.researchgate.net/publication/259948193_Predicting_Depression_via_Social_Media

Internet Slang Dataset

<https://floatcode.wordpress.com/2015/11/28/internet-slang-dataset/>

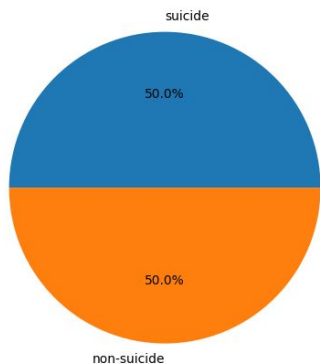
Inference from the papers cited:

- **Temporal Analysis:** Changes in behavior over time (e.g., frequency of negative posts) can signal escalating risk
[MDPI](#)
- **Contextual Features:** Surrounding circumstances, such as recent life events, may correlate with increased risk. External factors (like the content of replies or interactions) are also important
[MDPI](#)
- **Hybrid Models:** Traditional ML models like Perceptron, Logistic Regression and other bi-classification models can improve classification accuracy when extracting and analyzing different types of features (textual and linguistic)

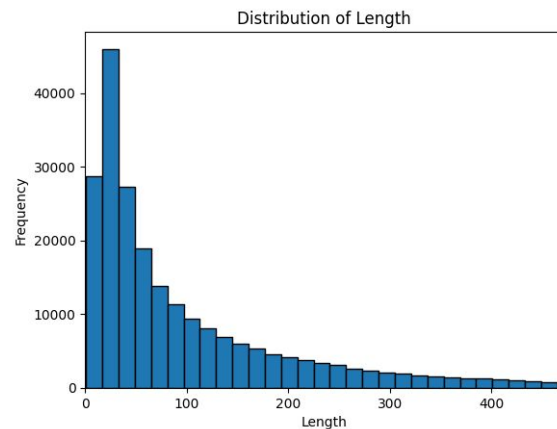
Dataset Description and EDA



- We utilized a publicly available Reddit dataset obtained from Kaggle, containing 2,30,000 posts made on the r/SuicideWatch between December 2008, and January 2021.
- Pie chart shows an equal distribution of posts in the dataset.
- A mean of 131.5 characters and a max of 15,632.
- The histogram visualizes the frequency distribution of post lengths, with most posts being under 100 characters.



```
count    232074.000000
mean      131.546438
std       222.081581
min        1.000000
25%       26.000000
50%       60.000000
75%      154.000000
max     15632.000000
Name: len, dtype: float64
```



Data preprocessing is essential for our Reddit dataset due to the following reasons:

1. Handling words and numbers that contribute to meaning and context
2. Removing Unnecessary Links
3. Dealing with Emojis

In summary, data preprocessing is crucial for ensuring the dataset is clean, consistent, and focused, especially since it was typed by various users. This step helps enhance the generalizability of our dataset, leading to more reliable analysis and outcomes.

This dataset comes from real social media posts and has been used in many research papers, which shows it is trustworthy. Because of its background and how it's been applied in studies, it is a good fit for detecting suicidal thoughts.

Data Preprocessing-

1. Convert all text to lowercase-Case differences are usually not relevant for meaning.
2. Remove URLs.
3. Remove numerical values and underscores Removing them simplifies the text.
4. Remove punctuation and special characters:Stripping them cleans the data for easier analysis.
5. Tokenize text
6. Remove stopwords: Common words like "the," "is," "for" don't add significant meaning to the text.
- 7.Lemmatization: reducing words to their base or root form

[Literature Behind the steps](#)

- We used feature extraction to simplify data, highlight significant terms and enhances model performance.
- It was also done to reduces noise, and improves interpretability, leading to more accurate identification of suicidal thoughts.
- This was done using Bag of Words and TF-IDF for feature selection.
- As they effectively capture important text features by representing word frequency and relevance.
- This approach enhances model performance by prioritizing significant terms, making it ideal for accurately detecting patterns in suicide-related content.

- 1. Logistic Regression-** Logistic Regression directly models the probability of a binary outcome using the sigmoid function, making it a natural choice for binary classification tasks.
- 2. Multinomial Naive Bayes-** Although designed for multi-class classification, it is effective for binary tasks when features are categorical.
- 3. Multilayer Perceptron (MLP)-**MLPs can model complex, non-linear relationships, making them powerful for binary classification tasks with intricate patterns in the data.
- 4. Decision Tree Classifier**
- 5. Random Forest-**An ensemble of decision trees, Random Forest reduces overfitting and handles non-linear relationships, providing robust performance for binary classification.
- 6. Support Vector Classifier (SVC)-**SVC excels at finding a decision boundary that maximizes class separation, making it particularly effective for binary classification, especially in high-dimensional spaces.

Data Preprocessing:

- A major problem we encountered was the concatenated words in the dataset which was treated as a single word by our model. To resolve this, we used ***wordninja[4]***, a Python Library, to split concatenated words into their individual components.
- Further we used ***spellchecker*** to split these concatenated words into proper English Words. However this was not very reliable as accuracy decreased after this. Hence LLM seem to be the best option. However free LLMs like google/flan-t5-small also performed poorly, paid LLM like GPT-4 or fine tuning of free ones seem to be the best option

Model Selection:

- We ran our Logistic Regression, Multinomial Naive Bayes and Perceptron model on the newly processed dataset and recorded the performance of each model.
- Then we tested Decision Tree, Random Forest and Multi Layer Perceptron models which are known for its highly accurate performance.

Results (Post Midsem)



- Model 1: Logistic Regression
- Model 2: Multinomial Naive Bayes
- Model 3: Perceptron
- Model 4: Decision Trees
- Model 5: Random Forest
- Model 6: Multi Layer Perceptron
- Model 7: Support Vector Machines

Model	Accuracy (%)
MultinomialNB	86.65
Perceptron	87.01
Logistic Regression	91.75
SVC	91.77
Decision Tree	84.19
Random Forest	89.36
Multi-Layer Perceptron	92.21

- **Best Performing Model:** Model 6 with the highest accuracy of (92.21%) and fully balanced precision, recall and F1-Scores.

Classification Report:				
	precision	recall	f1-score	support
non-suicide	0.92	0.92	0.92	23287
suicide	0.92	0.92	0.92	23128
accuracy			0.92	46415
macro avg	0.92	0.92	0.92	46415
weighted avg	0.92	0.92	0.92	46415

```
unseen_text = "i am feeling low for past few days i cant take it anymore"  
X_unseen = pd.Series([unseen_text])  
X_unseen_tfidf = tfidf.transform(X_unseen)  
print(model.predict(X_unseen_tfidf))
```

```
['suicide']
```

We explored various preprocessing steps like tfidf bag of words.wordninja, llm usage like google/flan-t5-small. But due to the decrease in the performance of models and LLM's[free LLM] inability to accurately identify the sentence's meaning, we decided to not include these methods. All the models perform reasonably well with MLP achieving the highest accuracy of 92.21%.

1. **Week 4:** (2nd September - 6th September) - Find and prepare a dataset for analysis.
2. **Week 5:** (9th September - 13th September) - Explore dataset for any outliers and try to clean noise out if any.
3. **Week 6-7:** (17th September - 30th September) - Experiment with different classification models.
4. **Week 8-9:** (14th October - 24th October) - Address data imbalances.
5. **Week 10-11:** (4th November - 14th November) - Test various regression models and apply them.
6. **Week 12-13:** (18th November - 29th November) - Combine outcomes for final analysis and conclusion

Individual Team Member's Contribution



■ Contribution in Project

